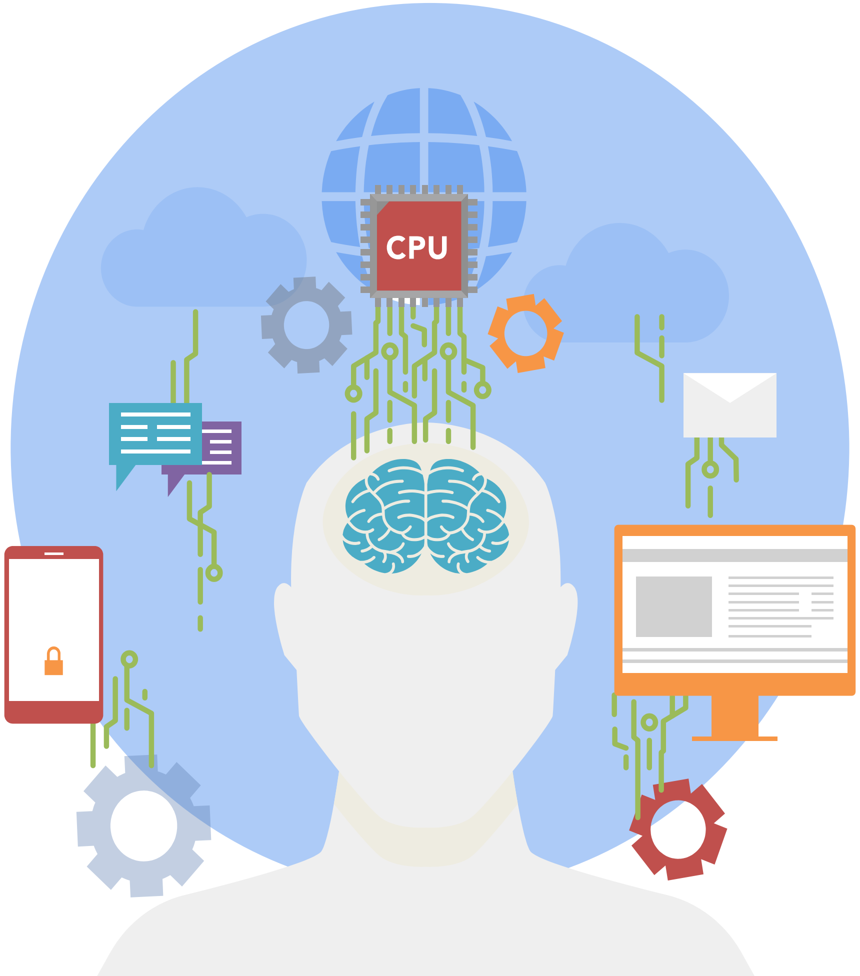


Understanding Agentic AI Part 2

Dr. Tamer Arafa

Patterns for highly autonomous agents

Planning workflows

Planning example: Customer service agent

Inventory Database

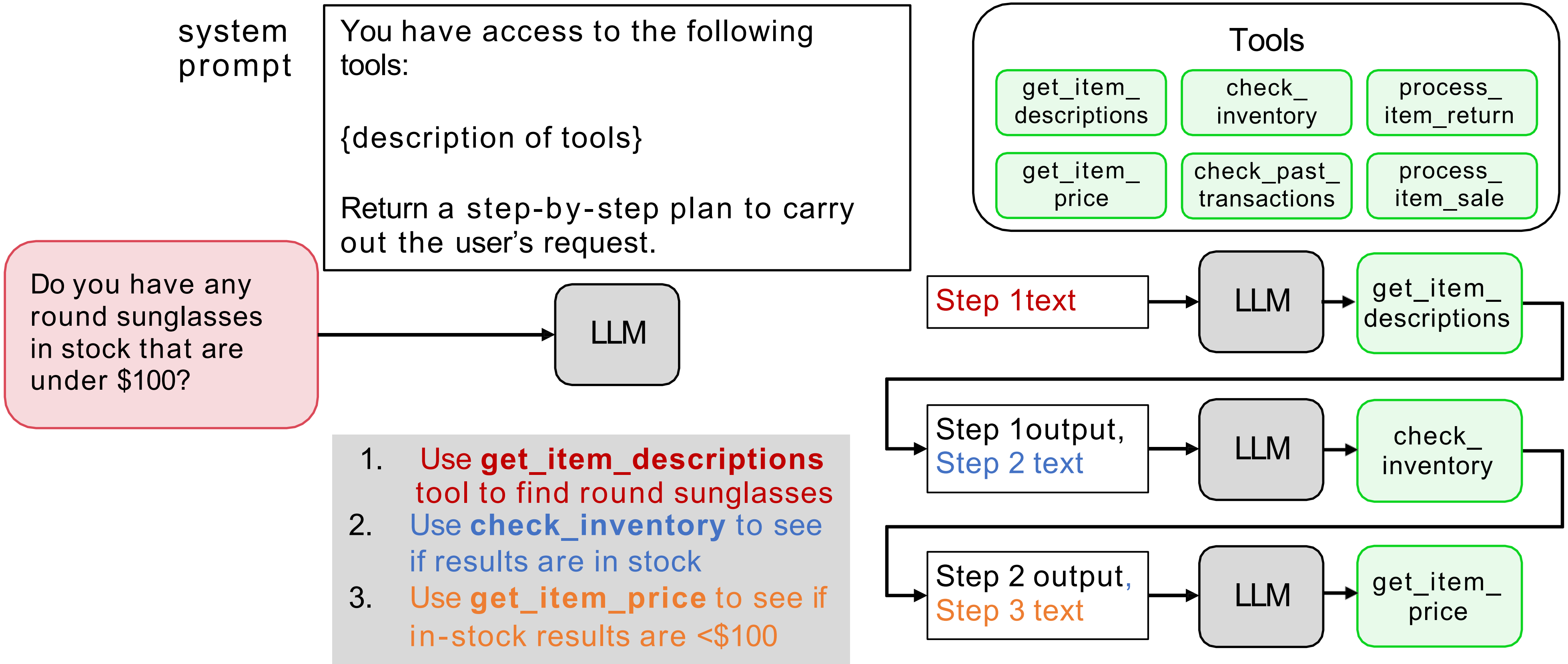
id	name	description	price	stock
1001	Aviator	Timeless pilot style for any occasion, metal frame	80	12
1002	Catseye	Glamorous 1950s profile, plastic frame	60	28
1003	Moon	Oversized round style, plastic frame	120	15
1004	Classic	Classic round profile, gold frame	60	9

Customer query:

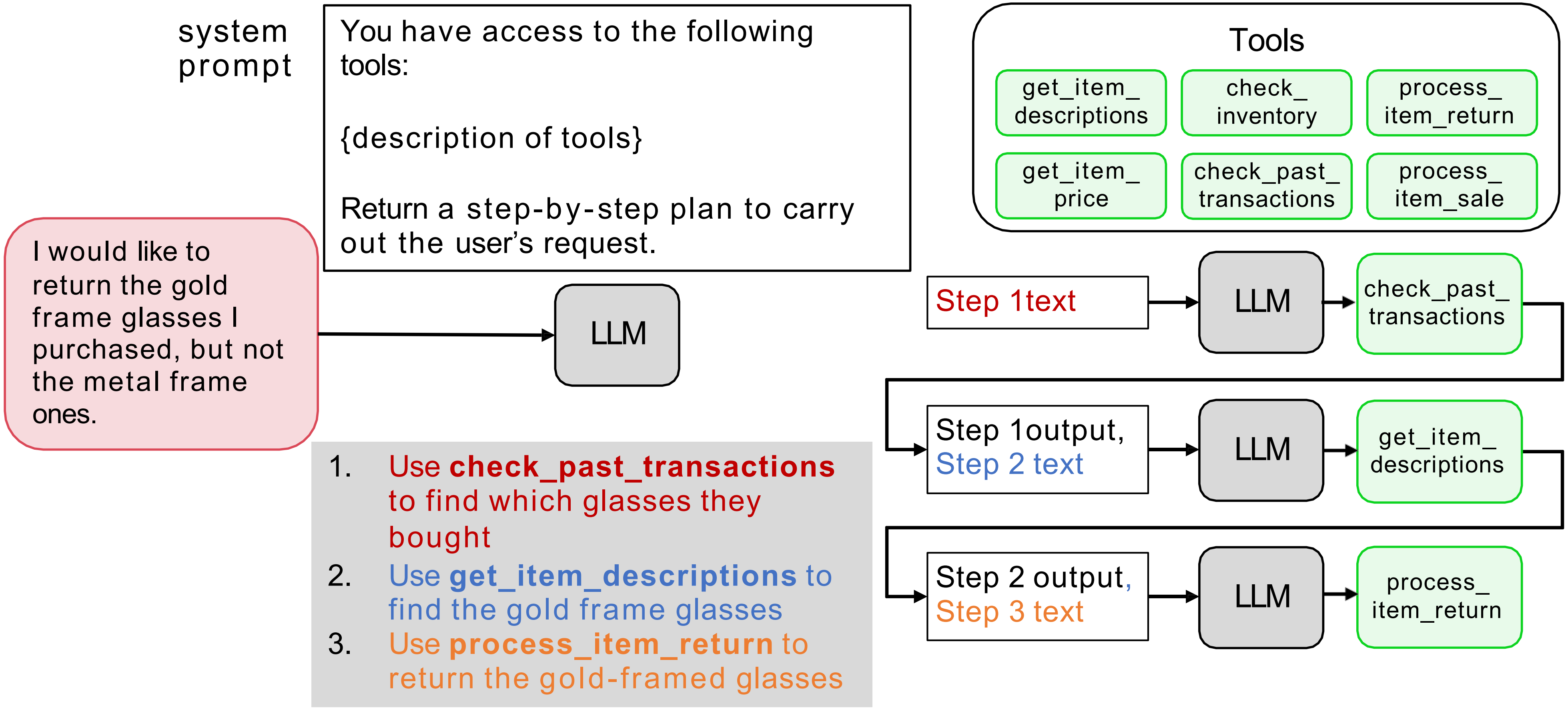
Do you have any round sunglasses in stock that are under $100?

Yes, we have our **Classic** sunglasses, which are a classic round metal frame and cost $60

Planning example: Customer service agent



Planning example: Customer service agent



Planning example: Email assistant

system prompt

You have access to the following tools:

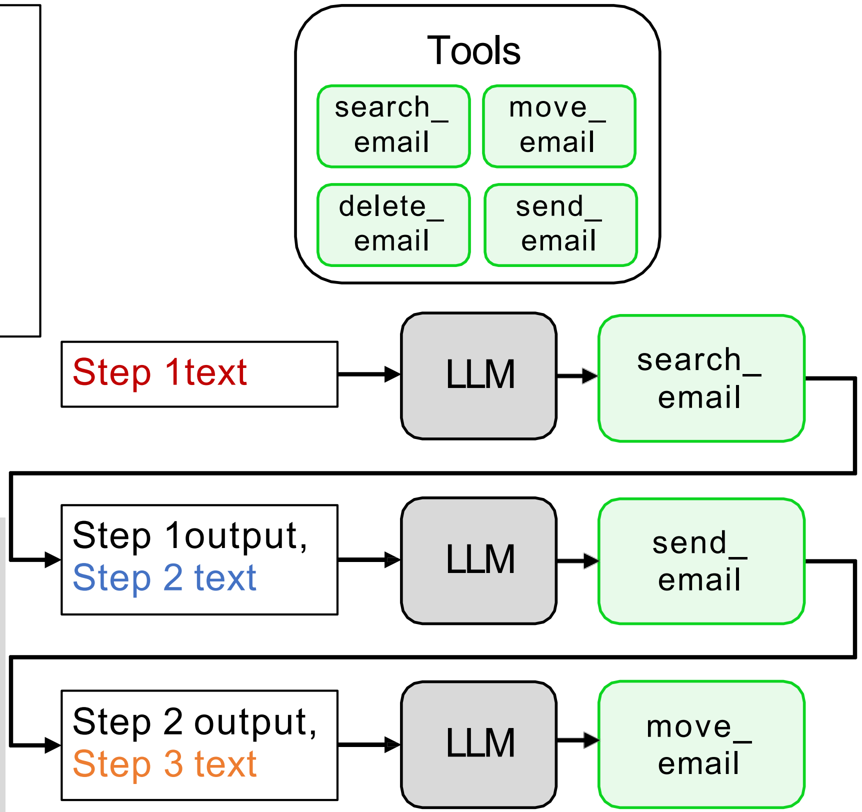
{description of tools}

Return a step-by-step plan to carry out the user's request.

Reply to that email invitation from Bob about dinner in New York and tell him I'll attend. Then archive his email.

LLM

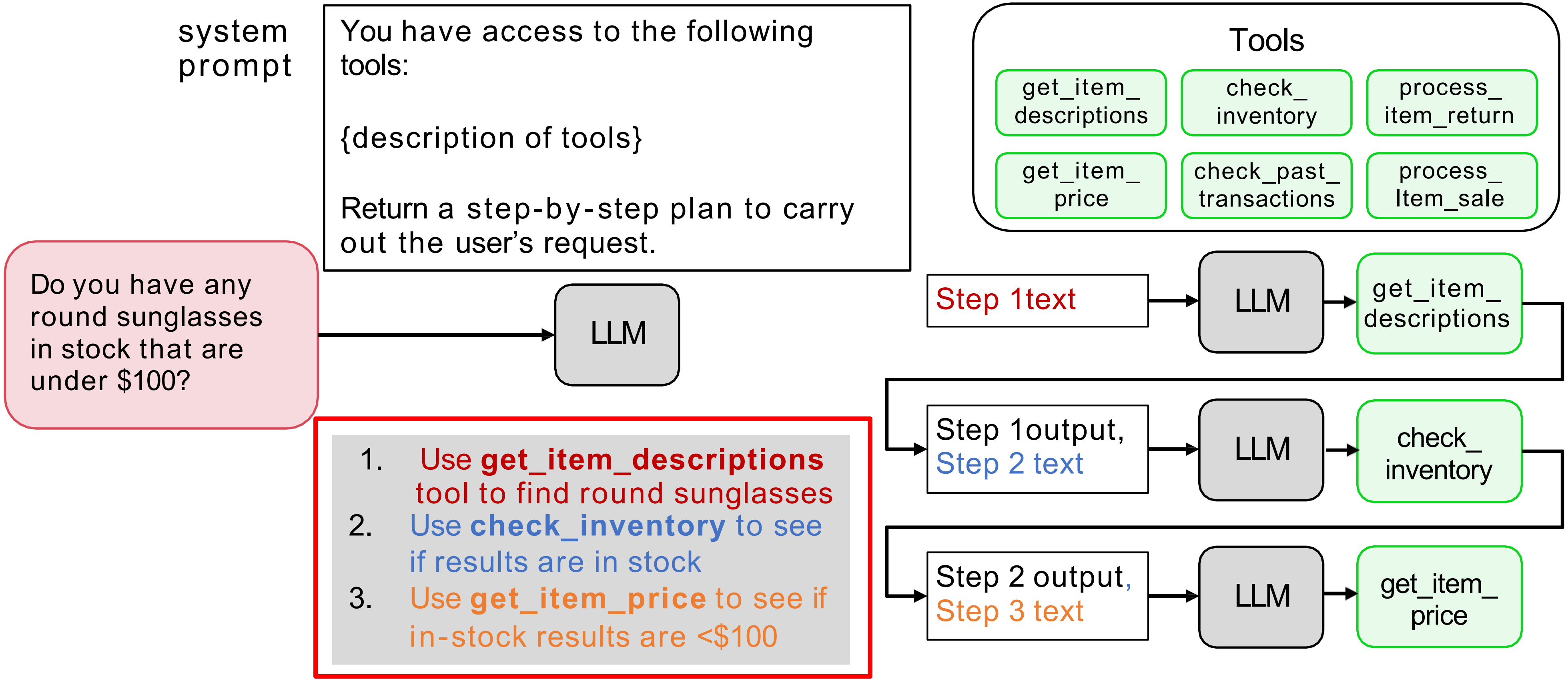
1. Use **search_email** to find emails from "Bob" that mention "dinner" and "New York"
2. Use **send_email** tool to reply and confirm attendance
3. Use **move_email** tool to move email to "archive" folder



Patterns for highly autonomous agents

Creating and executing LLM plans

Planning example: Customer service agent



Formatting plan as JSON

Updated system prompt

You have access to the following tools:

{description of tools}

Create a step-by-step plan in JSON format.

Each step should have the following items: step number, description, tool name, and args.

Do you have any round sunglasses in stock that are under $100?

→ LLM →

```
{
  "plan": [
    {
      "step": 1,
      "description": "Find round sunglasses",
      "tool": "get_item_descriptions",
      "args": {"query": "round sunglasses"}
    },
    {
      "step": 2,
      "description": "Check available stock",
      "tool": "check_inventory",
      "args": {"items": "results from step 1"}
    },
    ...
  ]
}
```

Patterns for highly autonomous agents

Planning with code execution

The challenge of planning with tools

Which month had the highest sales of hot chocolate? → LLM

Tools

get_column_max	get_column_mean	filter_rows
get_column_min	get_column_median	sum_rows

date	price	coffee_name	Size
2024-01-28	3.87	Hot Chocolate	M
2024-03-01	2.89	Cappuccino	S
2024-03-04	3.87	Latte	M
...	...	...	
2025-03-23	4.57	Latte	L

coffee_sales.csv

1. Use the **filter_rows** tool to extract transactions in January with coffee_name “Hot Chocolate”
2. Use the **get_column_mean** to find the average amount
3. Use the **filter_rows** tool to extract transactions in February with coffee_name “Hot Chocolate”
4. Use the **get_column_mean** to find the average amount
5. Repeat for March, April, May, ..,December
6. Determine the month with highest average using results of previous steps

The challenge of planning with tools

Which month had the highest sales of hot chocolate?

→ LLM

Tools

get_column_max | get_column_mean | filter_rows
get_column_min | get_column_median | sum_rows

date	price	coffee_name	Size
2024-01-28	3.87	Hot Chocolate	M
2024-03-01	2.89	Cappuccino	S
2024-03-04	3.87	Latte	M
...	...	...	
2025-03-23	4.57	Latte	L

How many unique transactions last week?

get_unique_entries

What were the amounts of the last 5 transactions?

get_last_N_values

... ...

- Brittle
- Inefficient
- Continuously dealing with edge cases

Planning with code execution

system prompt

Write code to solve the user's query. Return your answer as python code delimited with <execute_python> and </execute_python> tags.

What were the amounts of the last 5 transactions?

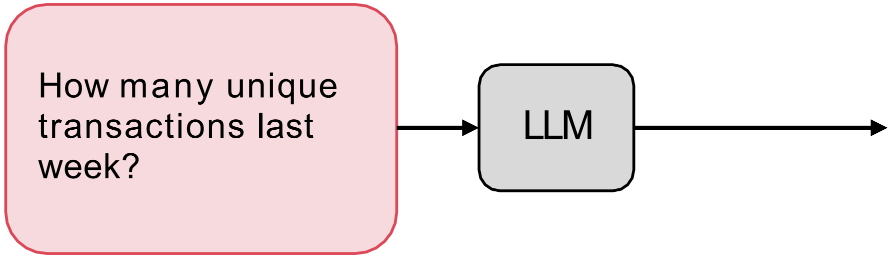
LLM

```
<execute_python>

import pandas as pd

# Load CSV
df = pd.read_csv("transactions.csv")

# Ensure date column is parsed as datetime
df["date"] = pd.to_datetime(df["date"])

# Sort by date to get most recent transactions
df_sorted = df.sort_values(by="date",
ascending=False)


# Select the last 5 transactions
last_5 = df_sorted.head(5)

# Show just the price column (amounts)
print("Last 5 transaction amounts:")
print(last_5["price"].to_list())

</execute_python>
```

Planning with code execution

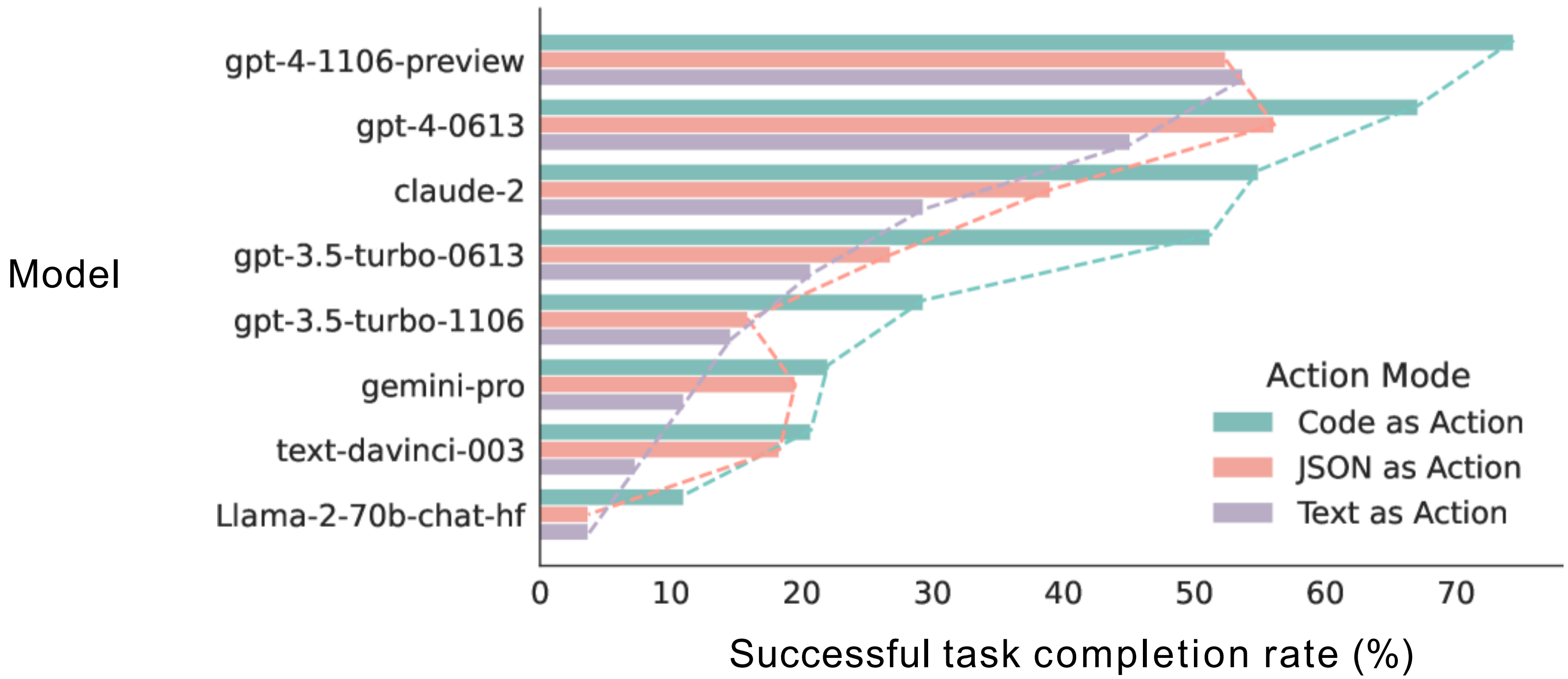
system prompt

Write code to solve the user's query. Return your answer as python code delimited with <execute_python> and </execute_python> tags.




```
<execute_python>

import pandas as pd

# Read CSV and parse the "date" column as datetime
df = pd.read_csv("transactions.csv",
parse_dates=["date"])

# Define time window
today = pd.Timestamp.today()
week_ago = today - pd.Timedelta(days=7)

# Filter rows where date is within last week
last_week = df[df["date"].between(week_ago, today)]

# Drop duplicate rows and count
print(last_week.drop_duplicates().shape[0])

</execute_python>
```

Planning with code improves performance



[Adapted from “Executable Code actions Elicit Better LLM Agents”, Wang et al. 2024]

Patterns for highly autonomous agents

Multi-agentic workflows

Some tasks require more than 1person!

Task	Team
Create marketing assets	Researcher Graphic Designer Writer
Writing a research article	Researcher Statistician Lead writer Editor
Preparing a legal case	Associate Paralegal Investigator

Example: Marketing team

Researcher

Tasks
- Analyze market trends
- Research competitors

Tools
- Web search

researcher

Graphic designer

Tasks
- Create data visualizations
- Create artwork

Tools
- Image generation, manipulation
- Code execution for chart generation

graphic designer

Writer

Tasks
- Transform research into report text and marketing copy

Tools
- (None)

writer

Example: Marketing team with linear plan

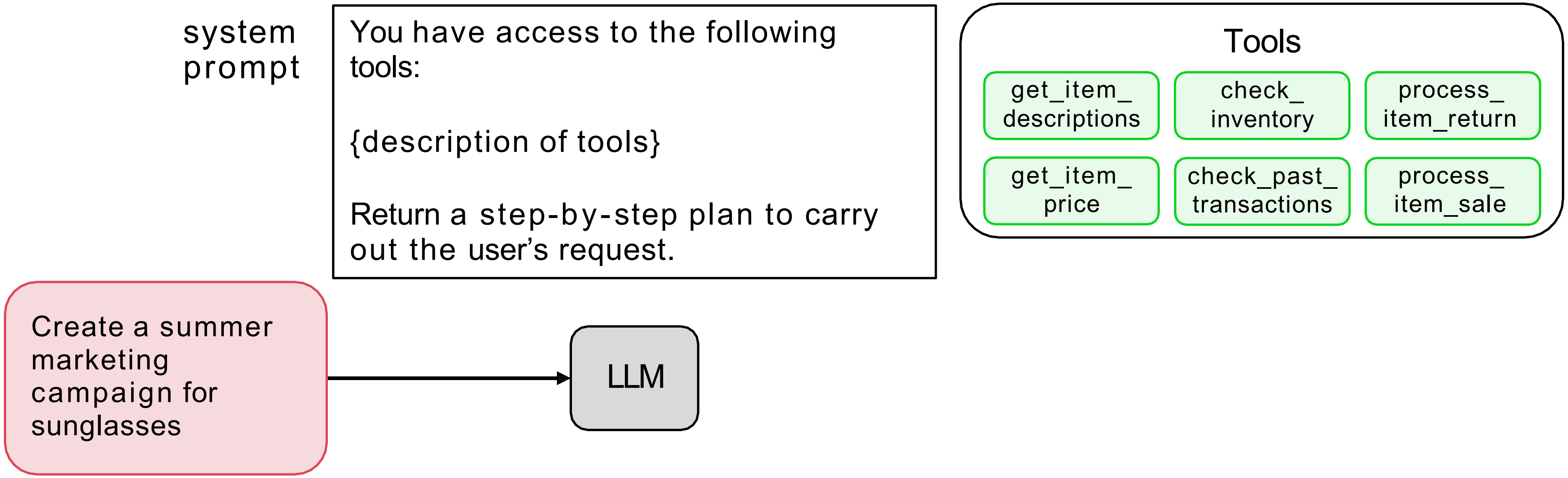
Create a summer marketing campaign for sunglasses → researcher → research → graphic designer → research, visualizations, artworks → writer

Here are current sunglasses trends and competitor offerings…

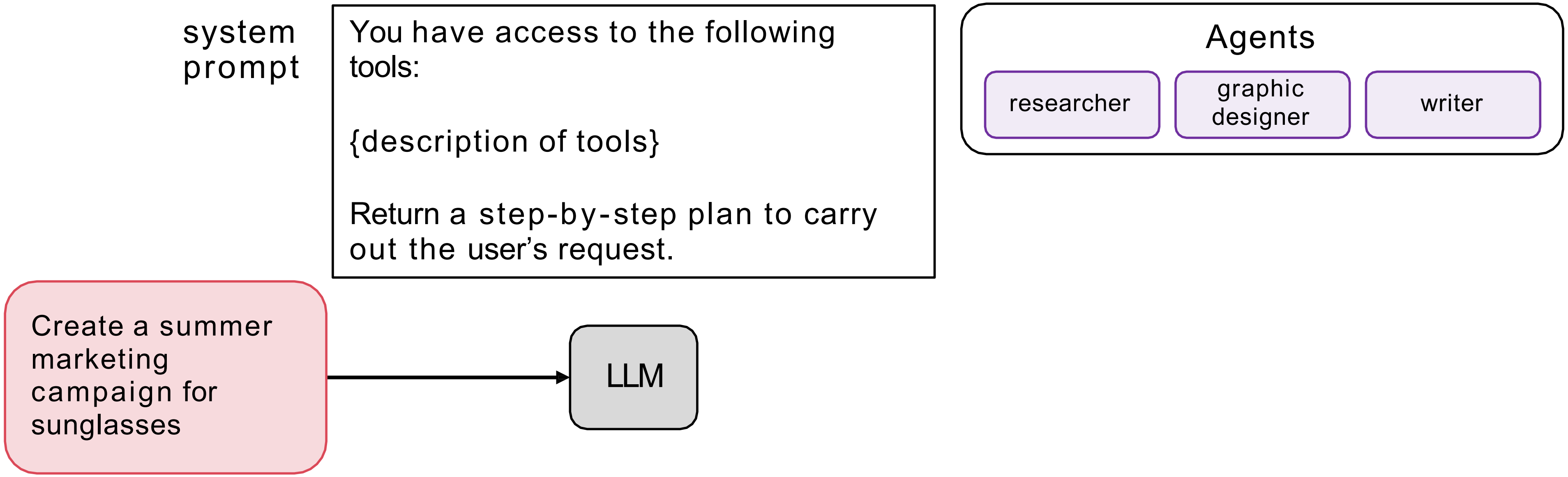
Here are 5 data visualizations and 5 artwork options for the report…

Final report

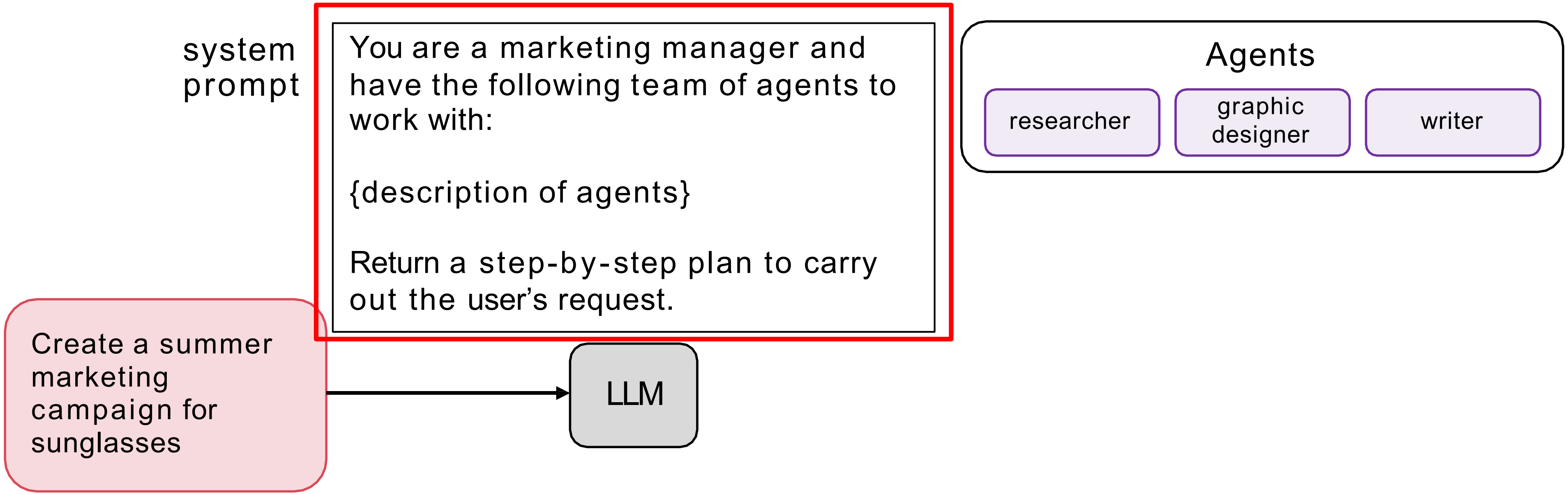
Example: Planning with multiple agents



Example: Planning with multiple agents



Example: Planning with multiple agents



Example: Planning with multiple agents

system prompt

You are a marketing manager and have the following team of agents to work with:

{description of agents}

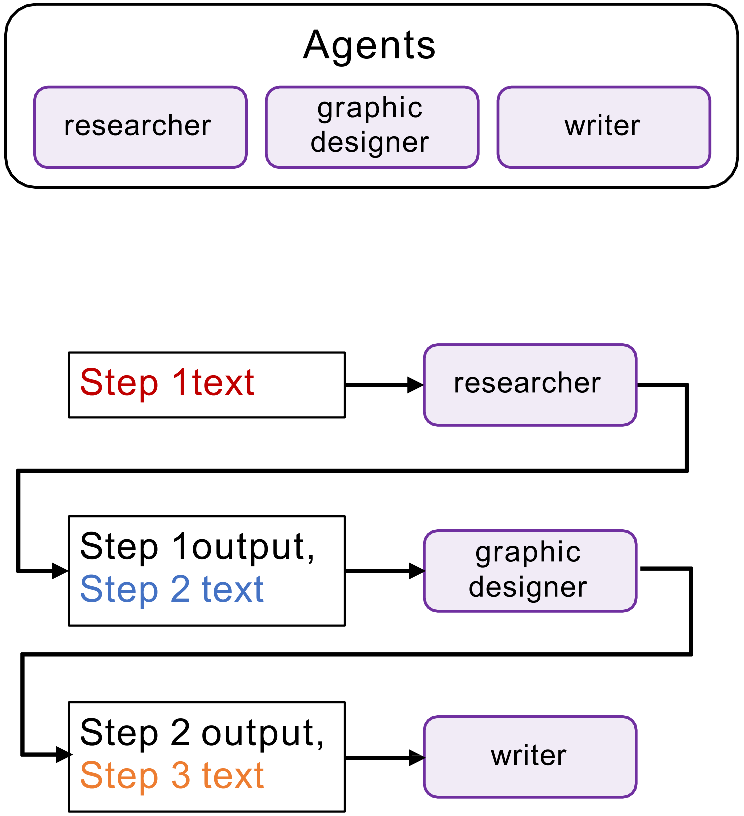
Return a step-by-step plan to carry out the user's request.

Create a summer marketing campaign for sunglasses

LLM

marketing manager

1. Ask **researcher** to research current sunglasses trends
2. Ask **graphic designer** to create ad images
3. Ask **writer** to create report
4. Review report



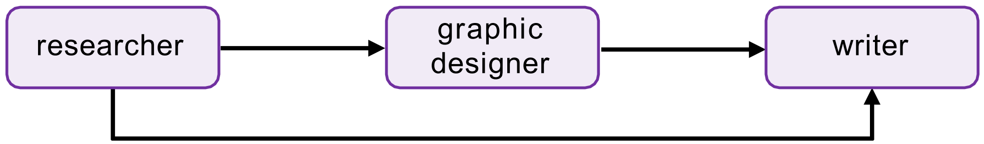
Patterns for highly autonomous agents

Communication patterns for multi-agent systems

Example: Marketing team with linear plan

Create a sum
marketing
campaign for
sunglasses

Communication Pattern



writer

Final report

Example: Planning with multiple agents

system prompt

You are a marketing manager and have the following team of agents to

Agents

graphic

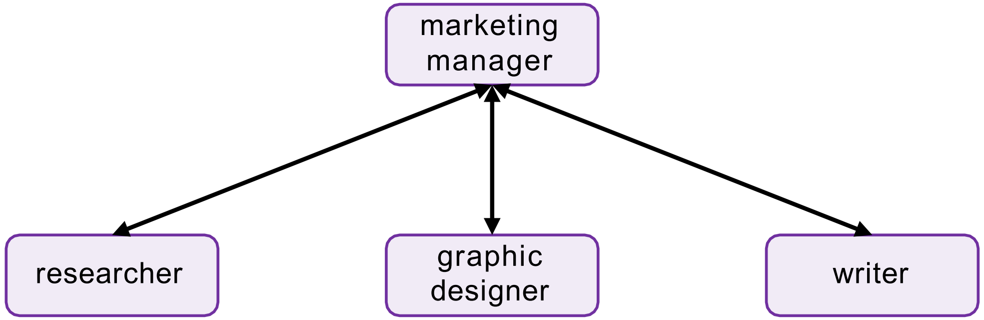
writer

Create a summ marketing campaign for sunglasses

archer

phic igner

Communication Pattern



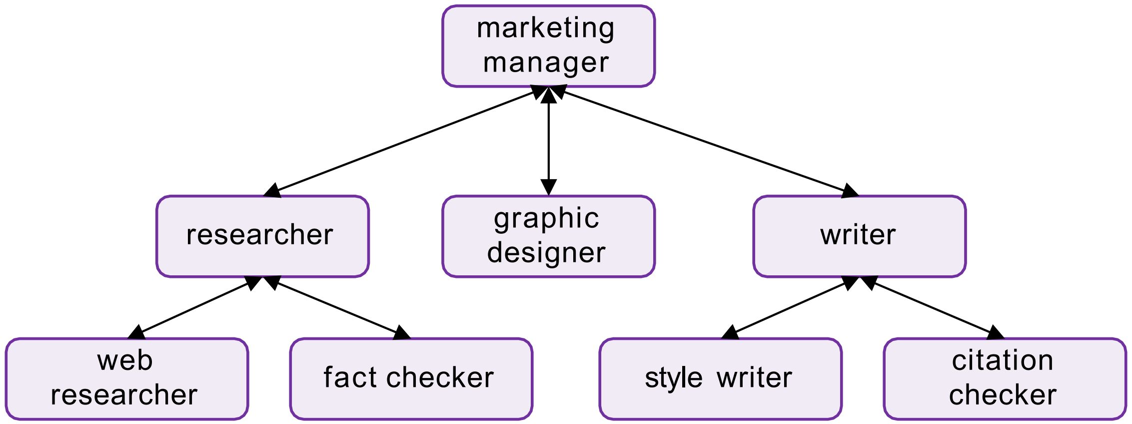
3. Ask **writer** to create report
4. Review report

Step 2 output,
Step 3 text

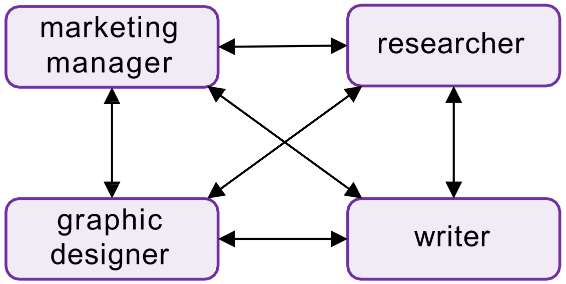
writer

Other communication patterns

Deeper hierarchy



All-to-all



Summary

- Why Agentic AI
- Reflection design pattern
- Tool use (function calling)
- Evals, error analysis
- Planning, multi-agent systems

Practical Tips for Building Agentic AI

Evaluations (evals)

Example: invoice processing workflow

TechFlow Solutions LLC

890 Juniper Drive
San Mateo, CA 94401
Phone: (415) 555-7890
Email: billing@techflowsol.com

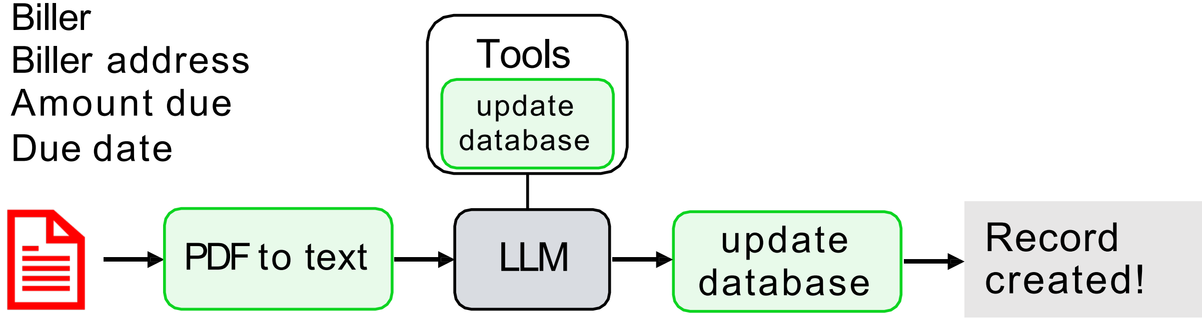
Due Date: August 20, 2025 **Invoice Date: August 6, 2025**

Description	**Qty**	**Unit Price**	**Line Total**
Consulting - Systems Integration (hrs)	20	$150.00	$3,000.00

Total Due: **$3,000.00**

4 required fields:

Biller
Biller address
Amount due
Due date



10-20 invoices

Create an eval to measure date extraction

1. Manually extract due dates from 10-20 invoices

test invoice 1 per example ground truth

"August 20, 2025" → "2025/08/20"

2. Specify output format of data in prompt

Format the due date as YYYY/MM/DD

3. Extract date from the LLM response using code

```
date_pattern = r'\d{4}/\d{2}/\d{2}'
extracted_date = re.findall(date_pattern, llm_response)
```

4. Compare LLM result to ground truth

```
if (extracted_date == actual_date):
    num_correct +=1
```

Driving your development process with evals

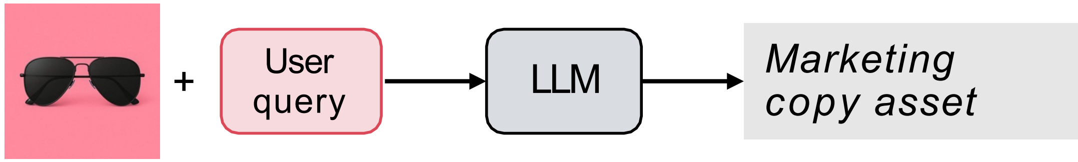
- Build a system and look at outputs to discover where it is behaving in an unsatisfactory way
  - E.g. incorrect due dates in invoice data extract

- Drive improvement by putting in place a small eval with ~20 examples to help you track progress

- Monitor as you make changes to workflow (e.g. new prompts, new algorithms) and see if the metric improves

Example: marketing copy assistant

Length guidelines:
Instagram caption: 10 words max



	17 words
	Ok
	Ok
	14 words
	11 words

Create an eval to measure text length

1. Create a set of 10-20 test tasks

Image	Example prompt
	Create an Instagram post
	Create an Instagram post

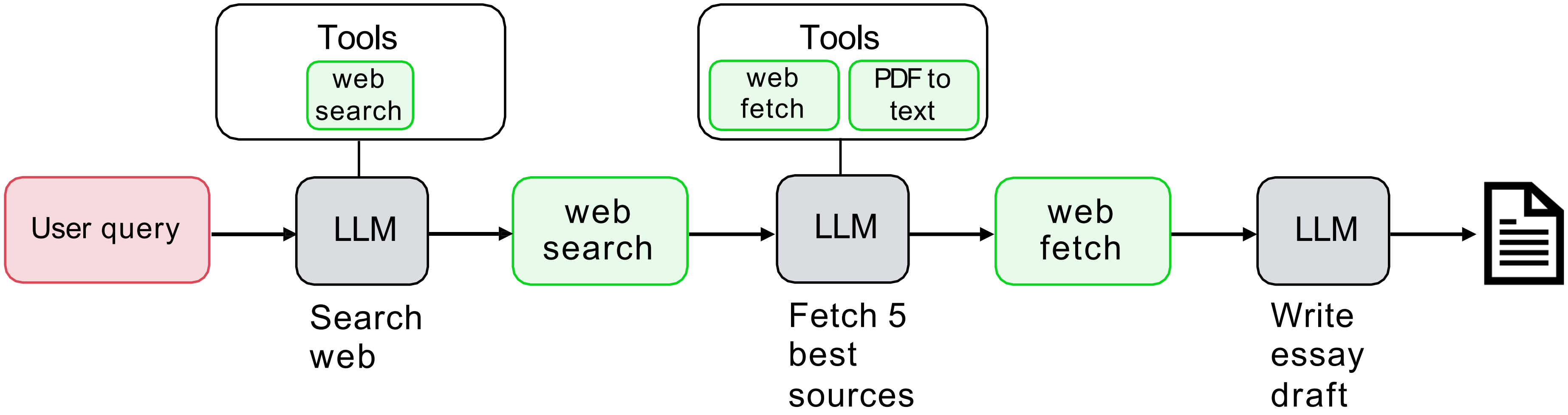
2. Add code to measure word count of the output

```
word_count = len(text.split())
```

3. Compare length of generated text to limit

```
if (word_count <= 10):
    num_correct +=1
```

Example: research agent



Prompt	Issues
Recent black hole science	Missed high-profile result that had lots of news coverage
Renting vs buying a home in Seattle?	Seems to do a good job
Robotics for harvesting fruit	Didn't mention leading equipment company

Sometimes misses points a human would have made

Create an eval to measure performance

1. Choose 3-5 gold standard discussion points for each topic

Example prompt	Gold-standard talking points
Black holes	Event horizon, radio telescope
Robotic harvesting	RoboPick, pinchers

ground truth annotations

2. Use LLM-as-a-judge to count how many topics were mentioned

Determine how many of the 5 gold-standard talking points are present in the provided essay.
Original Prompt
{original_prompt}
Essay to Evaluate
{essay_text}
Gold Standard Talking Points
{gold_standard_points}
Output Format
Return a json object with two keys: score (a single number between 0 and 5), and explanation (a string that lists the talking points present)

3. Get score for each prompt in eval set

Two "axes" of evaluation

	Evaluate with code (objective)	LLM-as-judge (subjective)

Per example ground truth	Checking invoice date extraction
	`if (extracted_date == actual_date):`
	`    num_correct +=1`
		Counting gold-standard talking points
		Count the number of gold standard points in the following text…

No per example ground truth	Checking marketing copy length
	`if len(text) <= 10:`
	`    num_correct += 1`
		Grading charts with a rubric
		Grade this chart according to (i) whether it has clear axes labels, (ii) ….

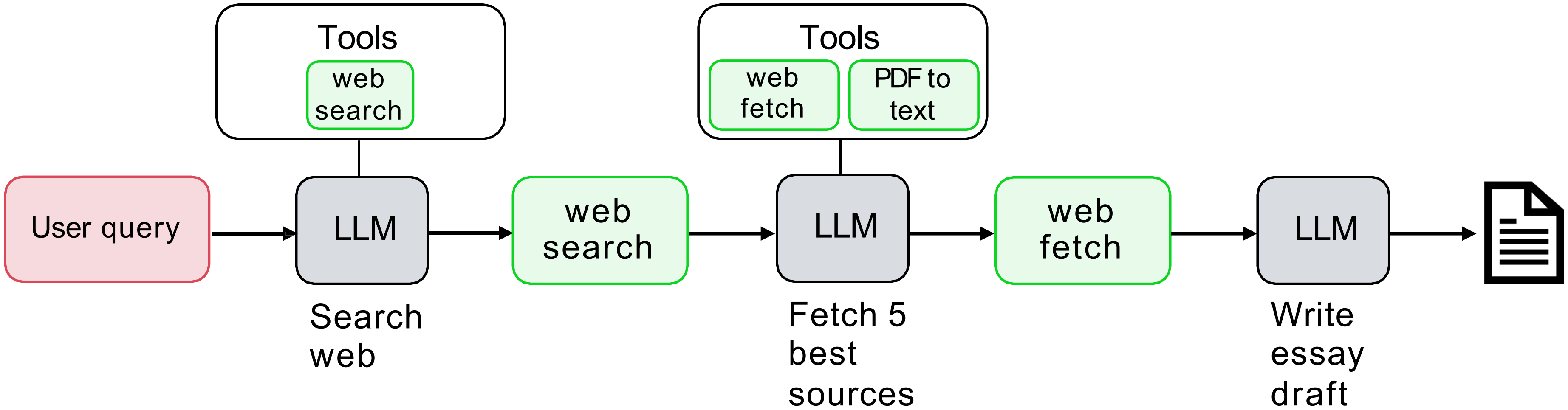
Tips for designing end-to-end evals

- Quick and dirty is ok to start!
- As you find places where your evals fail to capture human judgement as to what system is better, use that as an opportunity to improve the metric
- Look for places where performance is worse than humans

Practical Tips for Building Agentic AI

Error Analysis and prioritizing next steps

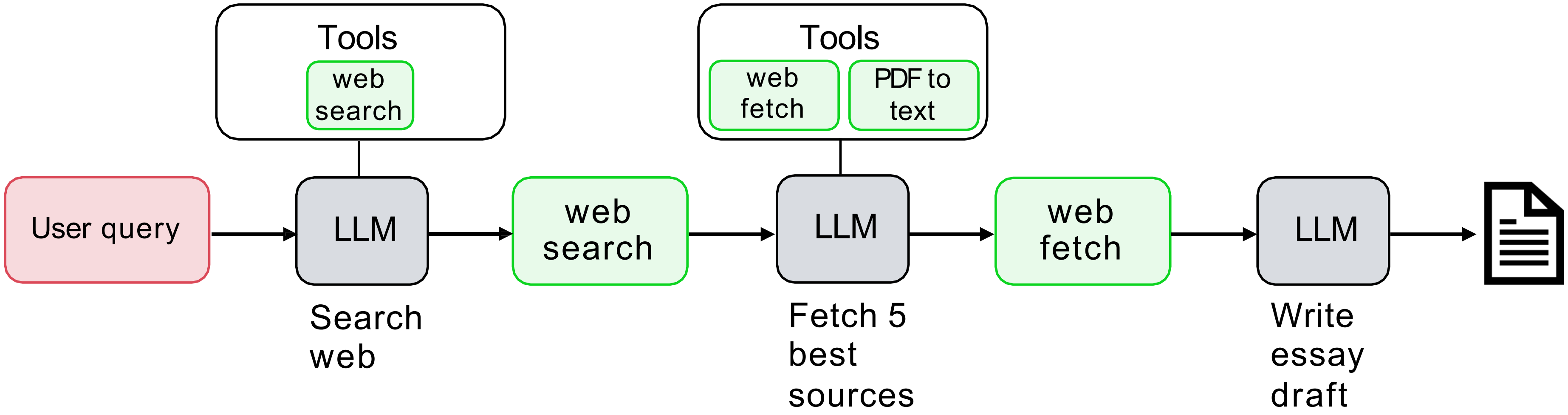
Example: research agent



Prompt	Issues
Recent black hole science	Missed high-profile result that had lots of news coverage
Renting vs buying a home in Seattle?	Seems to do a good job
Robotics for harvesting fruit	Didn't mention leading equipment company

Observed error mode: Sometimes misses key points a human would make

Example: research agent



Possible causes

Bad search terms?

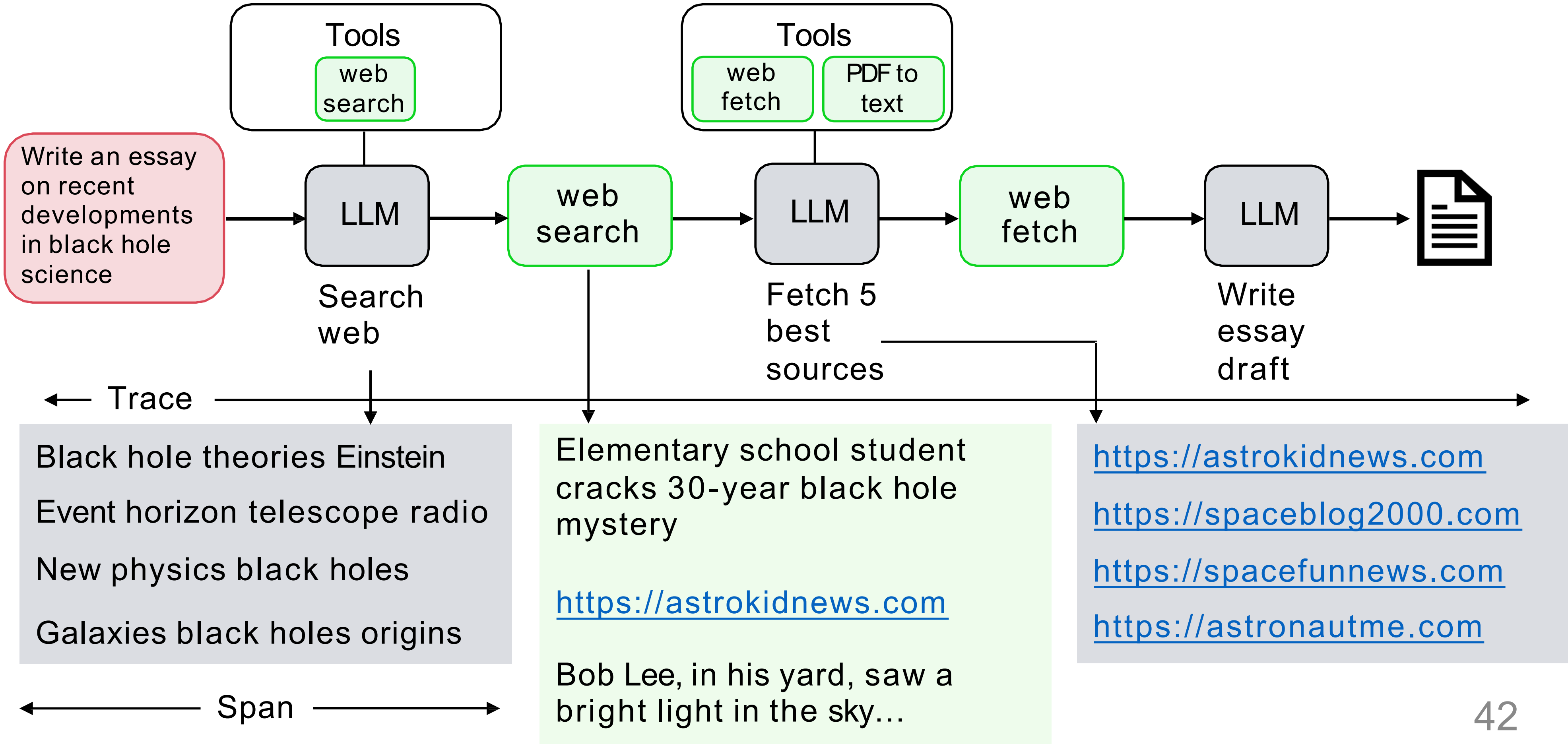
Low quality search results?

Poor selection of sources?

Bad reasoning over texts?

Examine traces to better understand each step in the workflow

Looking at traces



Counting up the errors

Prompt	Search terms	Search results	Picking 5 best sources	…	…
Recent developments in black hole science		Too many blog posts, not enough papers			
Renting vs buying a home in Seattle			Missed well-known blog		
Robotics for harvesting fruit	Terms too generic	Website for elementary school students			
…	…	…	…		
Batteries for electric vehicles		Only selected US-based companies	Missed magazine		
	5%	**45%**	**10%**	…	…

Tips for error analysis

- Develop a habit of looking at traces
- Carry out error analysis to figure out what component performed poorly, leading to a poor final output
- Use error analysis output to decide where to focus efforts

Practical Tips for Building Agentic AI

More error analysis examples

Example: Invoice processing workflow

TechFlow Solutions LLC

890 Juniper Drive
San Mateo, CA 94401
Phone: (415) 555-7890
Email: billing@techflowsol.com

Due Date: August 20, 2025 **Invoice Date: August 6, 2025**

Description	Qty	Unit Price	Line Total
Consulting - Systems Integration (hrs)	20	$150.00	$3,000.00

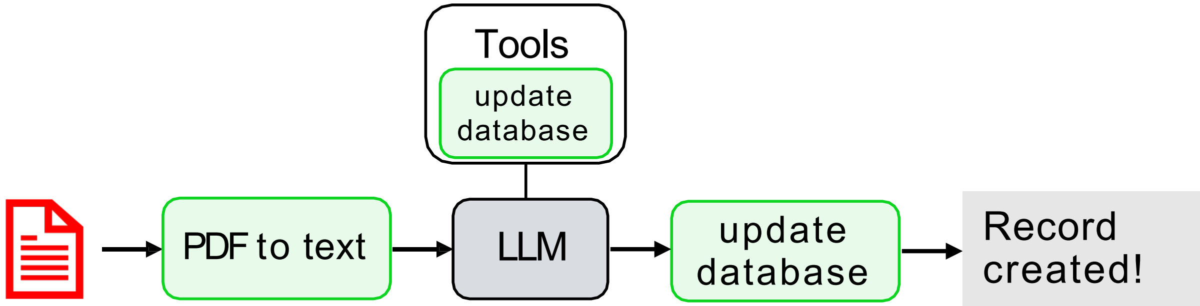
Total Due: **$3,000.00**

<u>4 required fields:</u>

Biller
Biller address
Amount due
Due date

Steps:

1. Identify required fields
2. Record in database



To carry out error analysis, focus on examples where performance is subpar

Counting up the errors

- Select 10-100 invoices for which the agentic workflow extracted the wrong due date

Input	PDF-to-text	LLM data extraction
Invoice 1	Errors in extraction	
Invoice 2		Wrong date selected
Invoice 3		Wrong data selected
...	...	...
Invoice 20	Errors in extraction	Wrong data selected
	15%	87%

Example: Responding to customer email

From: Susan Jones
Subject: Wrong item shipped

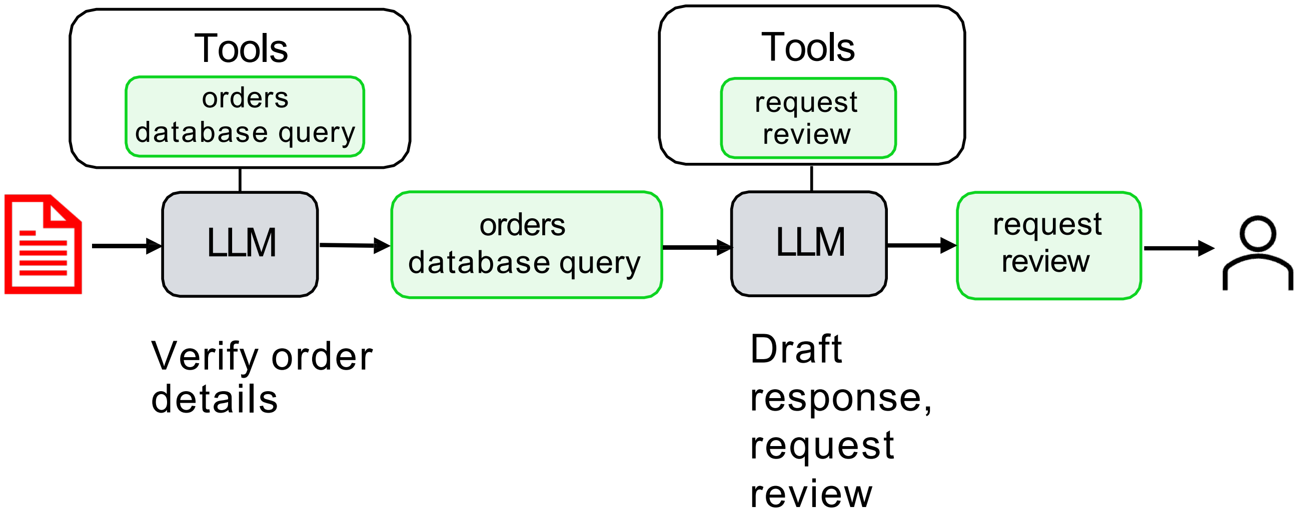
I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan

Steps:

1. Extract key information
2. Find relevant customer records
3. Draft response for human review



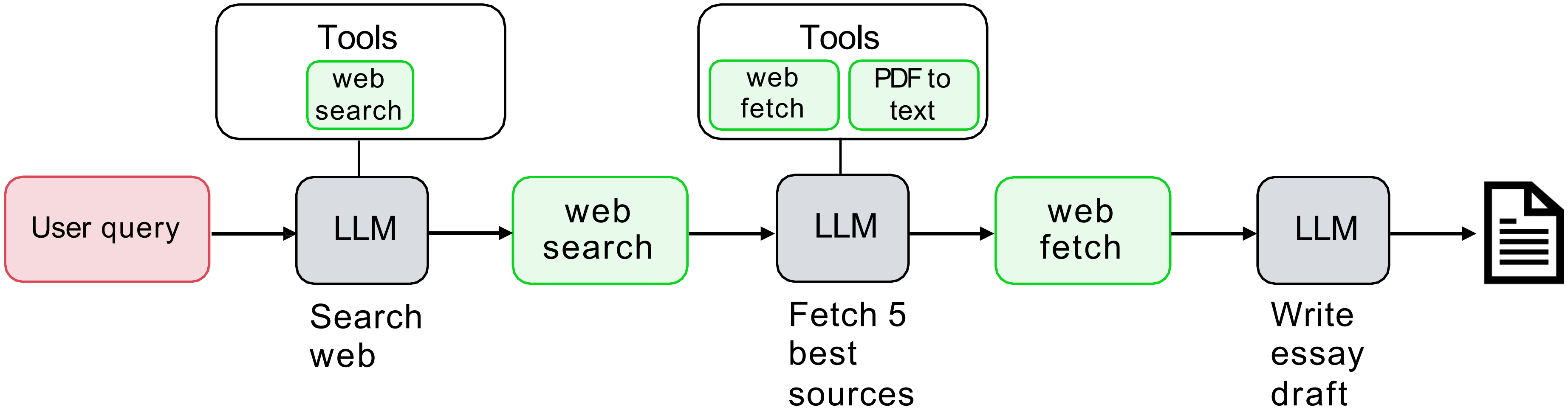
Counting up the errors

Input	LLM-drafted query	Orders database query	LLM-drafted email
Email 1	Wrong table		
Email 2		Error in database entry	Didn't address details of order
Email 3	Incorrect math		
...	...	...	...
Email 50			Defensive tone
	75%	4%	30%

Practical Tips for Building Agentic AI

Component-level evaluations

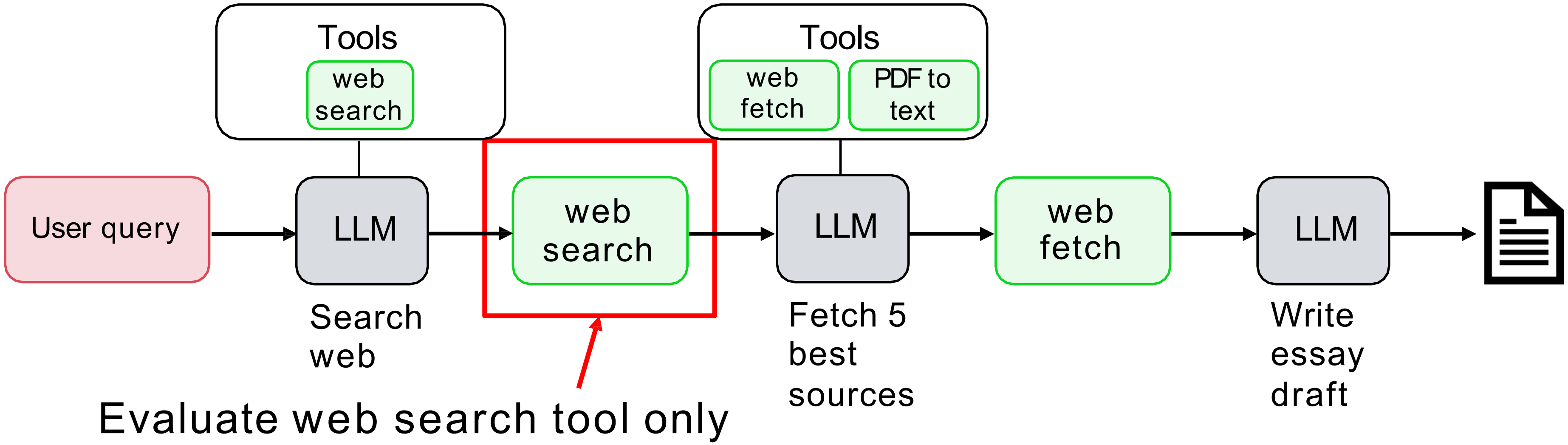
Example: research agent



Prompt	Issues
Recent black hole science	Missed high-profile result that had lots of news coverage
Renting vs buying a home in Seattle?	Seems to do a good job
Robotics for harvesting fruit	Didn't mention leading equipment company

End-to-end eval is expensive!

Example: research agent



- Create a list of gold standard web resources
- Write code that calculates how many results correspond to gold standard websites e.g. F1-score
- Track as you vary hyperparameters: e.g., search engine, number of results, dates

Benefits of component-level evaluations

- Can provide clearer signal for specific errors
  - Avoid the noise in end-to-end system
- More efficient for focused team to optimize
  - Work on smaller, more targeted problems faster

Practical Tips for Building Agentic AI

How to address problems you identify

Improving non-LLM component performance

E.g. web search, text retrieval for RAG, code execution, trained ML model (for speech recognition, people detection, etc.)

• Tune hyperparameters of component	Web search: Number of results, date range RAG: Change similarity threshold, chunk size ML models: Detection threshold
• Replace the component	Try a different web search engine, RAG provider, etc.

Improving LLM component performance

• Improve your prompts	Add more explicit instructions. Add one or more concrete example to the prompt (few-shot prompting)
• Try a new model	Try multiple LLMs and use evals to pick the best
• Split up the step	Decompose the task into smaller steps
• Fine-tune a model	Fine tune on your internal data to improve performance

Instruction following

Summary of customer call:

On July 14, 2023, Jessica Alvarez (SSN: 555-44-3333) of 1024 Maple Ridge Lane, Boulder, CO 80301, submitted a support ticket....

Prompt

> Identify all cases of personally identifiable information (PII) in the text below.
>
> Then return a list of the identified PII classified by type, and then redact all the identified PII with "*****".
>
> Separate the list and the redacted text with "REDACTED: ".
> {text}

Instruction following results (Llama 3.1 8B)

The identified personally identifiable information (PII) classified by type is:

- Social Security Number (SSN): 524-18-7629
- Address: 1024 Maple Ridge Lane, Boulder, CO 80301

Redacted text with "******" replacing the identified PII:

On July 14, 2023, ***** (SSN: *****), residing at *****, Boulder, CO 80301, submitted a support ticket…

List of identified PII classified by type
- SSN: 524-18-7629
- Address: 1024 Maple Ridge Lane, Boulder, CO 80301

Didn't follow the instructions properly

Missed some PII

Instruction following results (GPT-5)

Identified PII (type → value):
1. Full Name → Jessica T. Alvarez
2. Social Security Number → 524-18-7629
3. Physical Address → 1024 Maple Ridge Lane, Boulder, CO 80301

REDACTED:
On July 14, 2023, ***** (SSN: *****), residing at *****, submitted a support ticket...

Followed formatting instruction

Identified all PII

Developing intuition for model intelligence

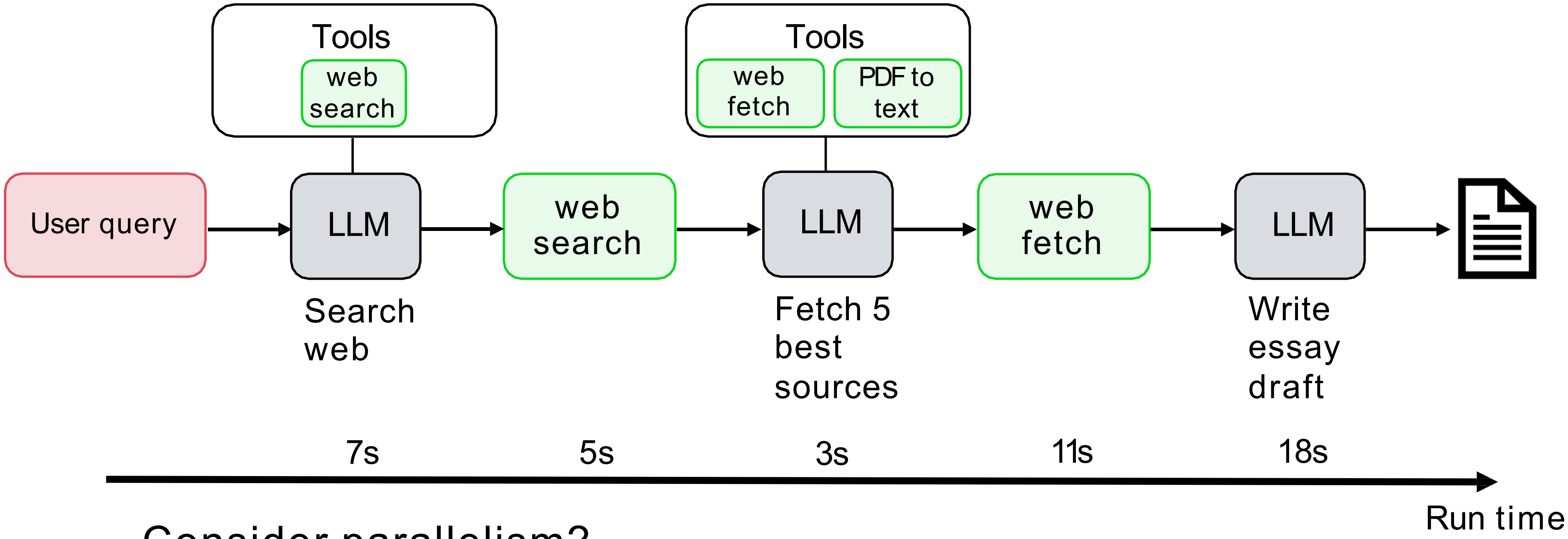
- Play with models often
  - Having a personal set of evals might be helpful
  - Read other people’s prompts for ideas of how to best use models

- Use different models in your agentic workflows
  - Which models work for which types of tasks?
  - aisuite makes it easy to quickly swap out models

Practical Tips for Building Agentic AI

Latency, cost optimization

Example: research agent



Consider parallelism?

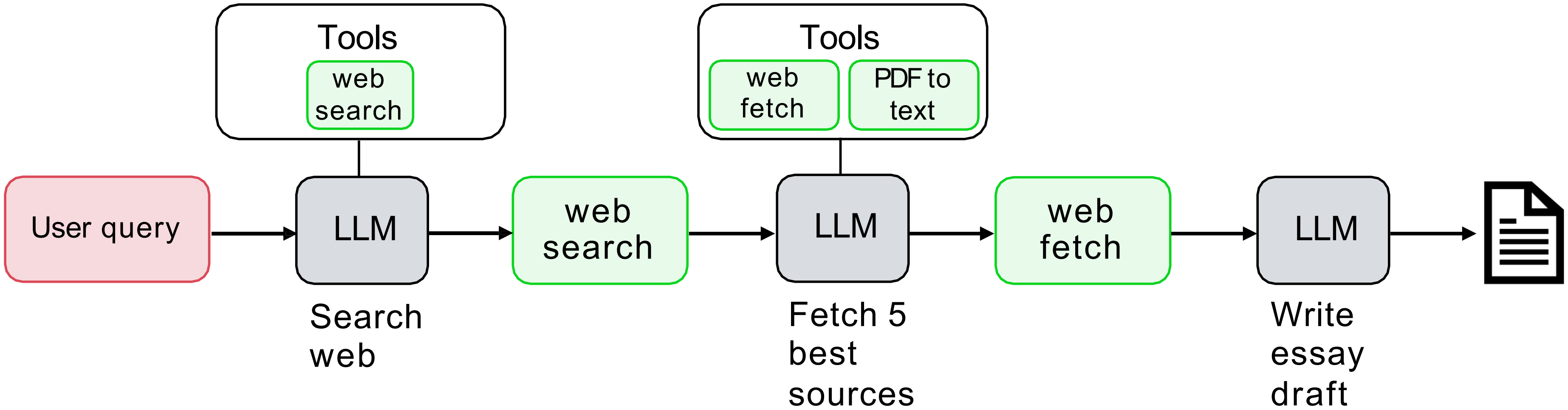
LLM steps too long?

- Try smaller/less intelligent model, or faster LLM provider

Costing your workflow

- LLM steps (pay per token)
- Any API-calling tools (pay per API call)
- Compute steps (based on server capacity/cost)

Example: research agent



Tokens: $0.0004

API call: $0.016

Tokens: $0.0004

API call: $0.002

PDF-to-text: $0.03

Tokens: $0.009

Practical Tips for Building Agentic AI

Development process summary

Development process summary

Build

Build end-to-end system

Improve individual component

Analyze

Examine outputs; traces

Build evals; compute metrics

Error analysis

Component-level evals

Lecture References

- Agentic AI – DeepLearning.AI Course (Modules 4-5)